

Uniform Estimates for Landau–de Gennes Minimizers with Point and Line Defects

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Joint work with Haotong Fu and Huaijie Wang

Academic Forum on Theoretical and Numerical Analysis
of Liquid Crystals and Complex Systems

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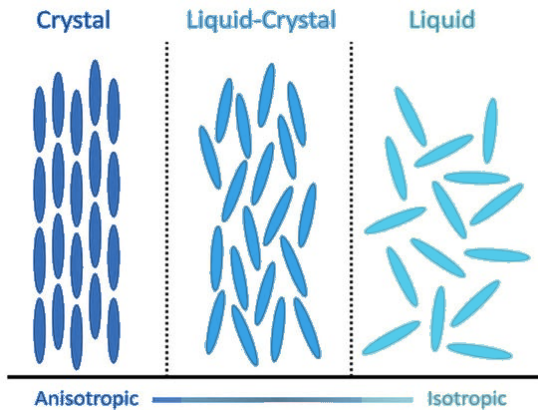
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Liquid Crystals

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Source: Auriau-Usson-Jouk, *J. Cardiovasc. Dev. Dis.* 2022, Fig. 1, CC BY 4.0.

Nematic, Smectic, and Cholesteric Phases

Key distinctions: orientational order and positional order.

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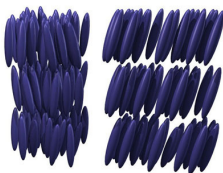
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Nematic



Smectic A / Smectic C



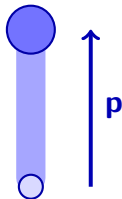
Cholesteric

Images: Kebes, Wikimedia Commons, CC BY-SA 3.0/GFDL

Vector Model (Oseen–Frank): One Local Orientation

Polar molecule

head and tail are different

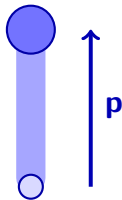


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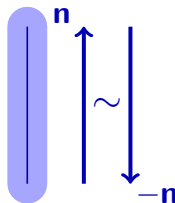
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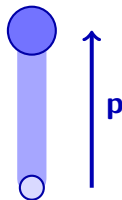


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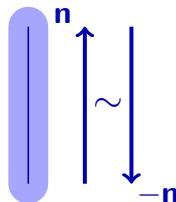
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The Oseen–Frank/vector model is simple and geometric, but rod-like nematics require the head-to-tail identification $\mathbf{n} \sim -\mathbf{n}$.

Molecular Model (Onsager): Orientation Distributions

Onsager idea: at each material point x , keep a probability distribution $f(x, \mathbf{m})$ of molecular axes.

$$\begin{aligned} f(x, \mathbf{m}) &\geq 0, \\ \int_{\mathbb{S}^2} f(x, \mathbf{m}) \, d\mathbf{m} &= 1. \end{aligned}$$

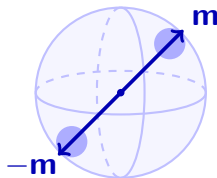
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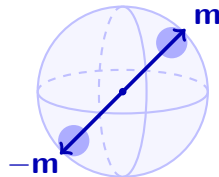
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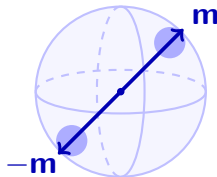
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- This captures orientational disorder, but the unknown now lives on $\Omega \times \mathbb{S}^2$.

Q-tensor Model (Landau–de Gennes)

Molecular model. At each point x ,

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Why not a vector? The first moment cancels:

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- $\mathbf{m} \otimes \mathbf{m}$ is unchanged when $\mathbf{m}$ is replaced by $-\mathbf{m}$.
- Subtracting $\frac{1}{3} \mathbf{I}$ removes the isotropic background and makes $\mathbf{Q}$ traceless.

Q-tensor Model: What Does $\mathbf{Q}$ Measure?

Natural state space

$$\mathbb{S}_0 := \{\mathbf{Q} \in \mathbb{M}^{3 \times 3} : \mathbf{Q}^T = \mathbf{Q}, \operatorname{tr} \mathbf{Q} = 0\}.$$

Two structural constraints

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where $\mathbf{n}_i \cdot \mathbf{n}_j = \delta_{ij}$ and $\lambda_1 + \lambda_2 + \lambda_3 = 0$.

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- **Eigenvectors** give preferred axes;
- **Eigenvalues** measure the strength of the bias along these axes.

Q-tensor Model: Isotropic, Uniaxial, and Biaxial States

Choose two orthonormal eigenvectors $\mathbf{n}, \mathbf{m}$ and let ℓ denote the third orthogonal eigenvector. Since $\text{tr } \mathbf{Q} = 0$, only **two scalar strengths** are independent:

$$\mathbf{Q} = s \left(\mathbf{n} \otimes \mathbf{n} - \frac{1}{3} \mathbf{I} \right) + r \left(\mathbf{m} \otimes \mathbf{m} - \frac{1}{3} \mathbf{I} \right), \quad s, r \in \mathbb{R}, \quad \mathbf{n} \cdot \mathbf{m} = 0.$$

In this basis the eigenvalues are

$$\lambda_{\mathbf{n}} = \frac{2s - r}{3}, \quad \lambda_{\mathbf{m}} = \frac{2r - s}{3}, \quad \lambda_{\ell} = -\frac{s + r}{3}.$$

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- **Biaxial state:** the three eigenvalues are distinct. Equivalently, s, r are both non-zero and $s \neq r$, so two independent axes are needed.

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Landau–de Gennes Free Energy

In the Landau–de Gennes theory, the state variable is a map $\mathbf{Q} : \Omega \rightarrow \mathbb{S}_0$. The free energy has two parts:

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$$f_e(\mathbf{Q}) = \frac{L_1}{2} |\nabla \mathbf{Q}|^2 + \frac{L_2}{2} \partial_j \mathbf{Q}_{ij} \partial_k \mathbf{Q}_{ik} + \frac{L_3}{2} \partial_k \mathbf{Q}_{ij} \partial_j \mathbf{Q}_{ik}.$$

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Bulk potential: the local free-energy density that selects the preferred degree and type of order.

$$f_b(\mathbf{Q}) = -\frac{a}{2} \operatorname{tr} \mathbf{Q}^2 - \frac{b}{3} \operatorname{tr} \mathbf{Q}^3 + \frac{c}{4} (\operatorname{tr} \mathbf{Q}^2)^2, \quad a \geq 0, \quad b, c > 0.$$

Equilibrium configurations are modeled by local minimizers of F .

Vanishing Elasticity Limit: Physical Scale

One-constant reduction. With $L_2 = L_3 = 0$ and after nondimensionalization, on a domain $\Omega \subset \mathbb{R}^3$, let

$$E_\varepsilon(\mathbf{Q}, \Omega) := \int_\Omega \left\{ \frac{1}{2} |\nabla \mathbf{Q}|^2 + \frac{1}{\varepsilon^2} f(\mathbf{Q}) \right\} dx, \quad (\text{LdG})$$

where

$$f(\mathbf{Q}) := k - \frac{a}{2} \operatorname{tr} \mathbf{Q}^2 - \frac{b}{3} \operatorname{tr} \mathbf{Q}^3 + \frac{c}{4} (\operatorname{tr} \mathbf{Q}^2)^2, \quad \mathbf{Q} : \Omega \rightarrow \mathbb{S}_0,$$

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$$\frac{1}{\varepsilon^2} = \frac{A_0 R^2}{L}, \quad \varepsilon = \frac{\sqrt{L/A_0}}{R} \ll 1.$$

Thus ε is small because, on a macroscopic sample, the bulk penalty is strong relative to the elastic coefficient.

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Thus ε is small because, on a macroscopic sample, the bulk penalty is strong relative to the elastic coefficient. This is the regime considered below. Sources: (Majumdar–Zarnescu, *ARMA* '10) for the elastic scale; (Gartland, *MMA* '18) for this nondimensional scaling.

Remark: Meaning of the Limit $\varepsilon \rightarrow 0^+$

A pointwise molecular interpretation would require

$$\mathbf{Q} = \int_{\mathbb{S}^2} \left(\mathbf{m} \otimes \mathbf{m} - \frac{1}{3} \mathbf{I} \right) f(\mathbf{m}) d\mathbf{m}, \quad -\frac{1}{3} \leq \lambda_i(\mathbf{Q}) \leq \frac{2}{3}.$$

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- The polynomial LdG model deliberately **relaxes this microscopic constraint** and lets $\mathbf{Q}_\varepsilon : \Omega \rightarrow \mathbb{S}_0$.
- For fixed ε , $\mathbf{Q}_\varepsilon(x)$ need not have a strict probability interpretation, especially where defects or transitions force the order parameter to change.

Remark: Meaning of the Limit $\varepsilon \rightarrow 0^+$

A pointwise molecular interpretation would require

$$\mathbf{Q} = \int_{\mathbb{S}^2} \left(\mathbf{m} \otimes \mathbf{m} - \frac{1}{3} \mathbf{I} \right) f(\mathbf{m}) d\mathbf{m}, \quad -\frac{1}{3} \leq \lambda_i(\mathbf{Q}) \leq \frac{2}{3}.$$

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- A stricter model can enforce the admissible eigenvalue range by a singular bulk potential (Ball–Majumdar, *Mol. Cryst. Liq. Cryst.* '10).

The Bulk Potential Selects the Constrained Target

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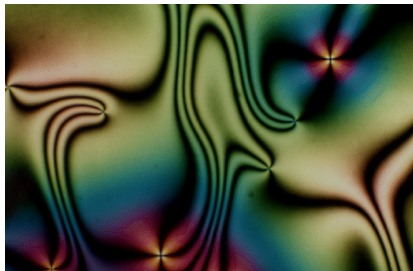
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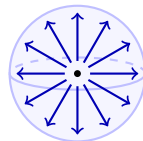
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Schlieren texture of a nematic film with surface point defects (boojums).
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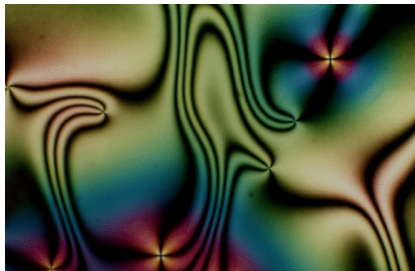


isolated core

A radial director **cannot be regular** at the center.

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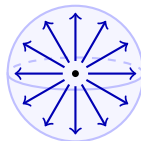
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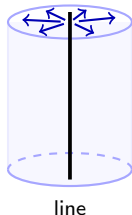
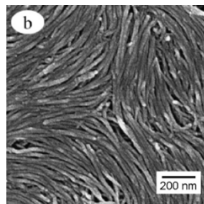
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$$\mathbf{n}(x) = \frac{x}{|x|}, \quad |\nabla \mathbf{n}(x)|^2 = \frac{2}{|x|^2},$$
$$\int_{B_1} |\nabla \mathbf{n}|^p dx \sim \int_0^1 r^{2-p} dr < +\infty \iff p < 3.$$

Point defects can have finite quadratic energy in three dimensions.

Line Defects Force a Logarithmic Scale

Physical picture. On a transverse disk, the identification $\mathbf{n} \sim -\mathbf{n}$ allows a half-turn after one full circuit around the core: this is an **index $\frac{1}{2}$ disclination**.



Transverse half-index model

$$\mathbf{x} = (r \cos \theta, r \sin \theta, z),$$

$$\mathbf{n}(\theta) = \left(\cos \frac{\theta}{2}, \sin \frac{\theta}{2}, 0 \right),$$

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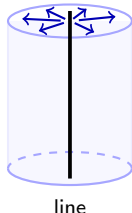
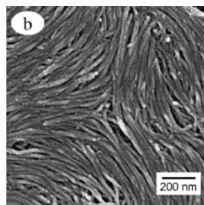
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Zhang/Kumar 2007

Carbon nano-tubes as liquid
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Two Energy Regimes

The constrained calculations separate two natural scales for the minimizers $\mathbf{Q}_\varepsilon$.

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Energy regime	Defect picture
Bounded energy: $E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega) = O(1)$	Point-defect behavior: isolated singularities may remain in the constrained limit
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Main question

Understand the defect sets, regularity, and convergence of minimizers for (LdG).

Previous Results: Bounded-Energy Regime

This regime is compatible with isolated point defects:

$$E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega) + \|\mathbf{Q}_\varepsilon\|_{L^\infty(\Omega)} \leq M.$$

The central question is whether $\mathbf{Q}_\varepsilon$ converges to the constrained harmonic-map picture away from the singular set.

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where $\mathbf{n}_0$ minimizes the Oseen–Frank director energy. Convergence is **uniform away from** **sing $\mathbf{Q}_0$** (the singular set of $\mathbf{Q}_0$).

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- **Flow analogue** (Wang–Wang–Zhang, *ARMA* '17): the relaxed $\mathbf{Q}$ -tensor flow converges, as $\varepsilon \rightarrow 0^+$, to the harmonic map flow in $\mathbb{R}^3$.

Previous Results: Logarithmic-Energy Regime

This regime is the natural scale for line defects:

$$E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega) \leq C |\log \varepsilon|, \quad \mu_\varepsilon := \frac{e_\varepsilon(\mathbf{Q}_\varepsilon)}{|\log \varepsilon|} dx.$$

The question is whether the limiting measure detects **one-dimensional defect sets**.

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- **Sextic bulk energy** (Wang–Zhang, preprint '24): the same point/logarithmic dichotomy is proved for sextic potentials, corresponding to the biaxial-phase setting.

Main Results: Bounded Energy

Theorem (Fu–Wang–Wang, preprint '25)

Suppose $\{\mathbf{Q}_\varepsilon\}_{\varepsilon \in (0,1)} \subset H_{\text{loc}}^1(\Omega, \mathbb{S}_0)$ are local minimizers of (LdG) such that

$$E_\varepsilon(\mathbf{Q}_\varepsilon, \Omega) + \|\mathbf{Q}_\varepsilon\|_{L^\infty(\Omega)} \leq M, \quad \forall \varepsilon \in (0, 1).$$

Then the following properties hold.

- 1 For any $p \in (1, 3)$, $\{\mathbf{Q}_\varepsilon\}_{\varepsilon \in (0,1)}$ is relatively compact in $W_{\text{loc}}^{1,p}(\Omega, \mathbb{S}_0)$.
- 2 For any $K \subset\subset \Omega$, there exists $C = C(a, b, c, K, M) > 0$ such that

$$\int_K \frac{1}{\varepsilon^2} f(\mathbf{Q}_\varepsilon) dx \leq C\varepsilon.$$

Main Results: Logarithmic Energy

Theorem (Fu–Wang–Wang, *SIAM J. Math. Anal.* '26)

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Then the following properties hold.

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- **Related targets:** the strategy is connected with Ginzburg–Landau-type targets and torus-like Landau–de Gennes solutions (Monteil–Rodiac–Van Schaftingen, *ARMA* '25; Dipasquale–Milot–Pisante, *ARMA* '21). The work (Canevari–Fu–Wang, preprint '26) extends the uniform estimates from the interior case to the global case.

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- **Anisotropic elastic terms.** When $L_2, L_3 \neq 0$, the analysis appears to require **additional structure**, for instance a small-anisotropy assumption.

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3 Sketch of the Proof

Bad Sets and Regular Scales

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Definition (Regular scales)

Let $\mathbf{Q} \in C^\infty(\Omega, \mathbb{S}_0)$. For $x \in \Omega$ and $\Lambda > 0$, define

$$\begin{aligned} r(\mathbf{Q}, x) &:= \sup\{r > 0 : r\|(|\nabla \mathbf{Q}| + r|D^2 \mathbf{Q}|)\|_{L^\infty(B_r(x))} \leq 1\}, \\ r^\Lambda(\mathbf{Q}, x) &:= \sup\{r > 0 : E_\varepsilon(\mathbf{Q}, B_r(x)) \leq \Lambda r\}. \end{aligned}$$

Bad Sets and Regular Scales

The proof turns regularity into scale estimates: a point is **bad** if either derivative control or energy control fails above scale r .

Definition (Regular scales)

Let $\mathbf{Q} \in C^\infty(\Omega, \mathbb{S}_0)$. For $x \in \Omega$ and $\Lambda > 0$, define

$$r(\mathbf{Q}, x) := \sup\{r > 0 : r\|(|\nabla \mathbf{Q}| + r|D^2 \mathbf{Q}|)\|_{L^\infty(B_r(x))} \leq 1\},$$
$$r^\Lambda(\mathbf{Q}, x) := \sup\{r > 0 : E_\varepsilon(\mathbf{Q}, B_r(x)) \leq \Lambda r\}.$$

Assume $\mathbf{Q}_\varepsilon \in H^1(\Omega, \mathbb{S}_0)$ is a local minimizer of (LdG). For $\Lambda, r > 0$, define

$$\text{Bad}_I(\mathbf{Q}_\varepsilon; r) := \{y \in \Omega : r(\mathbf{Q}_\varepsilon, y) < r\},$$
$$\text{Bad}_{II}(\mathbf{Q}_\varepsilon; r, \Lambda) := \{y \in \Omega : r^\Lambda(\mathbf{Q}_\varepsilon, y) < r\}.$$

- **Bad_I**: derivative regularity has not been reached by scale r .
- **Bad_{II}**: at some smaller scale, the energy is still too large relative to the line scale $E \sim r$.

Volume Estimates for Bad Sets

The main quantitative input is an estimate for neighborhoods of the bad sets:

$$\mathcal{L}^3(B_r(\text{Bad}_{\text{II}}(\mathbf{Q}_\varepsilon; r, \Lambda))), \quad \mathcal{L}^3(B_r(\text{Bad}_{\text{I}}(\mathbf{Q}_\varepsilon; r))).$$

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Bounded-energy regime

If $E_\varepsilon(\mathbf{Q}_\varepsilon, B_{40}) + \|\mathbf{Q}_\varepsilon\|_{L^\infty(B_{40})} \leq M$, then for suitable $\eta, \Lambda > 0$ and $\Lambda\varepsilon < r < 1$,

$$\mathcal{L}^3(B_r(\text{Bad}_{\text{I}}(\mathbf{Q}_\varepsilon; \eta r) \cap B_1)) \leq C(a, b, c, M)r^3.$$

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Logarithmic-energy regime

Let $\theta \in (0, 1)$. If

$$\frac{E_\varepsilon(\mathbf{Q}_\varepsilon, B_{40})}{|\log \varepsilon|} + \|\mathbf{Q}_\varepsilon\|_{L^\infty(B_{40})} \leq M$$

then for suitable $\eta, \Lambda > 0$, $\varepsilon^\theta < r < 1$, and $\varepsilon < \eta$,

$$\mathcal{L}^3(\text{Bad}_{II}(\mathbf{Q}_\varepsilon; r, \Lambda) \cap B_1) \leq C(a, b, c, M, \theta)r^2.$$

Sketch: From Bad Sets to Estimates

- **Type II bad set.** This estimate is based on [energy counting](#). If a point is in Bad_{II} , then on some ball B_ρ with $\rho > \varepsilon^\theta$, $\theta \in (0, 1)$,

$$E_\varepsilon(\mathbf{Q}_\varepsilon, B_\rho) \gtrsim \eta_0 \rho |\log \varepsilon|.$$

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- The proof is completed using arguments from fractional Allen–Cahn equations (Wang–Wei–Wu, preprint '25).

Sketch: Logarithmic Energy

The logarithmic regime has a genuine **line-defect scale**. The estimate for

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is weaker than the bounded-energy analogue, so the proof is split into two regions.

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Combining the two regions yields the uniform $W_{\text{loc}}^{1,p}$ estimate for every $1 < p < 2$ and the **uniform bulk-potential bound**.

Thank you for listening!